THE UNIVERSITY OF MELBOURNE

FINAL REPORT

# The Art of Scientific Computing:

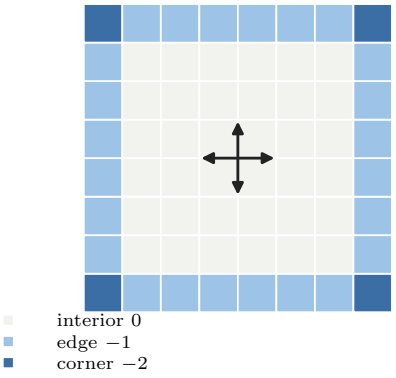
## Complex System:

## BTW Sandpile, with a Hexagonal Extension

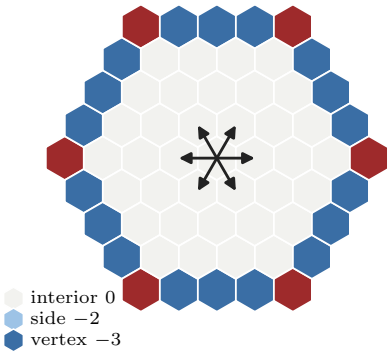

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***Subject Handbook code:***
**COMP-90072**

(a) square region, $z = 4$, $k = 4$




(b) hexagonal region, $z = 6$, $k = 6$





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# Contents

# Chapter 1

# Introduction

A small perturbation usually produces a small effect. Systems whose cause and effect are proportional are the ones we learn to solve first, because the response can be expanded about an equilibrium and the leading term is linear. The systems in this report are of the other kind.

Consider a pile of sand on a table. Grains are added one at a time, always at the same slow rate. Nothing about the driving is dramatic. Yet every so often a single grain lands and the whole pile rearranges: an *avalanche* involving anything from a handful of grains to a large fraction of the table, with no way to tell from the trigger which it will be. This is the behaviour Bak, Tang and Wiesenfeld identified as *self-organised criticality* (SOC) [**?**, **?**], and the sand pile they used to exhibit it is the model studied here.

## 1.1 Scale invariance and the power law

In an ordinary critical phenomenon, such as the liquid–gas transition, criticality is reached by tuning a control parameter to a special value, and correlations then extend over the whole system. SOC claims that certain systems reach such a state *by themselves*: they are slowly driven, the threshold for local rearrangement is fixed, and dissipation happens somewhere. No parameter is tuned.

The signature of that state is scale invariance. A function is scale invariant when the same shape reappears at every magnification, so that no size is singled out:

$$f(bx) = g(b)\, f(x) \qquad \text{for all } b > 0. \tag{1.1}$$

Putting $x = 1$ gives $g(b) = f(b)/f(1)$; substituting back, differentiating with respect to $b$ and setting $b = 1$ gives

$$x f'(x) = \alpha f(x), \qquad \alpha = \frac{f'(1)}{f(1)}, \qquad \Longrightarrow \qquad f(x) \propto x^{\alpha}. \tag{1.2}$$

A function with no preferred scale can only be a power of its argument, so scale invariance and power-law statistics are two descriptions of the same fact, and the argument requires only that $f$ be continuous, that $f(1) \neq 0$, and that a single $g$ serve every $b$ at fixed argument. The converse does not hold: a power law is compatible with mechanisms that are not self-similar, so a straight line on log–log axes is evidence for a candidate mechanism rather than a proof of one.

## 1.2 Non-equilibrium statistics

Three descriptions of a system that has stopped changing must be separated. A *steady state* requires only that macroscopic observables stop drifting with time. A *statistically stationary* state is stronger: the probability measure over configurations is time-independent, so while individual sites keep changing the distribution does not. *Equilibrium* is stronger again, requiring detailed balance and microscopic reversibility, so that every transition is as likely as its reverse.

The sand pile is statistically stationary but not in equilibrium. Sand moves one way — in at a random site, out at the boundary — so the dynamics is not time-reversible and detailed balance fails. The observable test is the input–output balance: in the stationary state the mean number of grains lost per drop is one. Chapter **??** develops the consequences of that measure.

This distinction matters for how results are read. At criticality the relative frequency of large and small avalanches is insensitive to the details of the local rule, so the quantities of interest are exponents and scaling functions, shared by whole families of models rather than computed from any one of them. That is what makes the extension in Chapter **??** a meaningful experiment: it changes a great many microscopic facts, and the question is which macroscopic ones survive.

## 1.3 Aim and structure

The project [**?**] asks for the BTW sand pile to be built, its SOC behaviour and non-Gaussian avalanche statistics studied, and the model then extended and compared with the baseline; twelve questions in the handbook guide the analysis.

Chapter **??** fixes the model. The toppling rule becomes a table $\Delta$, the four conditions on $\Delta$ are shown to be the whole content of conservation and loss, and the Abelian property is proved where the condition that carries it is stated. Chapter **??** describes the stationary state: the recurrent configurations, the exact mean height, and the distributions of the four avalanche observables. Chapter **??** turns the mathematics into a program. Chapter **??** replaces the square lattice by a hexagonal one and compares the two quantity by quantity — which quantities are universal, which are lattice-specific, and why the extension remains Abelian. Chapter **??** collects the conclusions. C++ types, function signatures and the matrix layer are in Appendix **??**.

# Chapter 2

# The model and the Abelian property

## 2.1 Sand on a grid, and the toppling rule as a table

Sand grains sit on an $L \times L$ square grid. The state at time $t$ is the height function

$$p_t(\boldsymbol{x}) \in \mathbb{N}_0, \qquad \boldsymbol{x} = (i, j) \in \boldsymbol{L} = \{1, \dots, L\}^2, \tag{2.1}$$

and the list of all $L^2$ heights is the *configuration*. Time does not advance during an avalanche: a drop is instantaneous and the whole rearrangement it triggers belongs to the same time step [**?**]. A configuration is *stable* when every site is below the threshold $k$, and *unstable* otherwise.

An unstable site gives one grain to each of its four nearest neighbours and loses four grains itself:

$$p(\boldsymbol{x}) \leftarrow p(\boldsymbol{x}) - 4, \qquad p(\boldsymbol{x} \pm \boldsymbol{e}_k) \leftarrow p(\boldsymbol{x} \pm \boldsymbol{e}_k) + 1, \quad k = 1, 2. \tag{2.2}$$

The rule has the four features required for self-organised criticality. The driving is slow: one grain per drop, at a rate far below the relaxation rate. The threshold is local: the test $p(\boldsymbol{x}) \geq k$ involves one site only. The redistribution is local: the grains reach the four neighbours and no other site. And the system dissipates: the grains that would leave the board are deleted. All four are properties of the rule rather than of a parameter value, so no quantity has to be tuned for the critical state to appear.

This changes five sites at once, which is awkward to reason about, so it is better recorded as how much each site $\boldsymbol{y}$ changes when $\boldsymbol{x}$ topples once. That table is the *toppling matrix*

$$\Delta(\boldsymbol{x}, \boldsymbol{y}) = \begin{cases} -4, & \boldsymbol{x} = \boldsymbol{y}, \\ 1, & \boldsymbol{x}, \boldsymbol{y} \text{ are nearest neighbours}, \\ 0, & \text{otherwise}, \end{cases} \tag{2.3}$$

and the rule is recovered by reading row $\boldsymbol{x}$: the diagonal entry is what the toppling site loses, the four non-zero off-diagonal entries are what its neighbours gain, and the zeros are the sites that do not notice. Summing row $\boldsymbol{x}$,

$$\sum_{\boldsymbol{y}} \Delta(\boldsymbol{x}, \boldsymbol{y}) = -4 + \#\{\text{neighbours of } \boldsymbol{x} \text{ inside the board}\}, \tag{2.4}$$

so the row sum reports conservation and loss in the same number: 0 for an interior site, $-1$ at an edge, $-2$ at a corner.

The row sum is negative exactly at the sites that have a neighbour outside the board, and nowhere else, so the boundary is the only exit for sand and the only reason the total stays bounded. Without it, on an infinite lattice or with grains that rebound instead of leaving, every row sum would be zero: the total sand would grow by one with every drop, no input–output balance would be possible, and both the total mass and the front of an avalanche would grow without bound, leaving no stationary state and therefore no exponents. The boundary does not have to be far away for this to work — any finite board will do — which is why an $L = 50$ board already shows the critical statistics.

### 2.1.1 The four conditions on $\Delta$

The handbook isolates four conditions for the dynamics to be well posed [**?**]; they are its equations (4a)–(4d), numbered here

$$\forall \boldsymbol{x} \neq \boldsymbol{y}: \quad \Delta(\boldsymbol{x}, \boldsymbol{y}) = \Delta(\boldsymbol{y}, \boldsymbol{x}) \geq 0, \tag{2.5a}$$

$$\forall \boldsymbol{x}: \quad \Delta(\boldsymbol{x}, \boldsymbol{x}) < 0, \tag{2.5b}$$

$$\forall \boldsymbol{x}: \quad \sum_{\boldsymbol{y}} \Delta(\boldsymbol{x}, \boldsymbol{y}) \leq 0, \tag{2.5c}$$

$$\sum_{\boldsymbol{x}, \boldsymbol{y}} \Delta(\boldsymbol{x}, \boldsymbol{y}) < 0. \tag{2.5d}$$

Each condition constrains one part of the dynamics. Condition (**??**) says sand moves between sites without a preferred direction: the non-negativity is what makes topplings commute (§**??**), and the symmetry makes $\Delta$ a real symmetric matrix, so that $-\Delta$ is a grounded graph Laplacian and $\det(-\Delta)$ counts the spanning trees of the wired graph by the matrix–tree theorem. Condition (**??**) says a toppling removes sand from the site that topples; without a strictly negative diagonal there would be no toppling at all, only redistribution, and the threshold would not be defined by the matrix. Condition (**??**) says a toppling never creates sand: it is (**??**) restated, and it is what bounds the total mass. Condition (**??**) says at least one place loses sand, which makes $-\Delta$ a nonsingular $M$-matrix and therefore makes $\Delta$ invertible, $\sigma$ unique by (**??**) and every avalanche finite (Proposition **??**).

All four hold for the matrix of (**??**), and they can be read off it directly. The off-diagonal entry is $1 \geq 0$ and the neighbour relation is symmetric, so (**??**) holds. The diagonal entry is $-4 < 0$, so (**??**) holds, and the threshold is fixed rather than chosen:

$$k = -\Delta(\boldsymbol{x}, \boldsymbol{x}) = 4. \tag{2.6}$$

The row sum of (**??**) takes the values 0, $-1$ and $-2$ for interior, edge and corner sites, so (**??**) holds; and since boundary rows are strictly negative while the board has a boundary, the total in (**??**) is negative. The threshold is a consequence of conservation, not a free parameter: if the rule sent a number of grains different from the number of neighbours, the interior row sum would no longer be zero and (**??**) would fail. That is the constraint which forces the threshold of the hexagonal lattice from 4 to 6 in Chapter **??**.

Note the convention: Dhar writes the rule as $z_{\boldsymbol{y}} \to z_{\boldsymbol{y}} - \Delta(\boldsymbol{x}, \boldsymbol{y})$ [**?**, **?**], so his $\Delta$ is the negative of ours and his conditions are these three with the sign flipped. The sign has to be checked whenever an exponent is taken from the literature.

### 2.1.2 The nine-site example

When $L = 3$ the board has nine sites and $\Delta$ is a $9 \times 9$ table. Numbering sites row-major, $(i, j) \mapsto 3i + j$, and filling row by row from (**??**):

$$\Delta = \begin{array}{c|ccccccccc} & 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ \hline 0 & -4 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & -4 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 2 & 0 & 1 & -4 & 0 & 0 & 1 & 0 & 0 & 0 \\ 3 & 1 & 0 & 0 & -4 & 1 & 0 & 1 & 0 & 0 \\ 4 & 0 & 1 & 0 & 1 & -4 & 1 & 0 & 1 & 0 \\ 5 & 0 & 0 & 1 & 0 & 1 & -4 & 0 & 0 & 1 \\ 6 & 0 & 0 & 0 & 1 & 0 & 0 & -4 & 1 & 0 \\ 7 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & -4 & 1 \\ 8 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & -4 \end{array} \tag{2.7}$$

with the site indices written outside. Row 4, the centre, is $(-4, +1, +1, +1, +1)$: the site takes $-4$ in column 4, its neighbours $+1$ in columns $1, 3, 5, 7$, the corners 0, and the sum is zero as (**??**) requires. Row 0 is a corner, with sum $-2$, so two grains leave the board whenever it topples.

### 2.1.3 Program: one local function

Equation (**??**) is a local function of constant cost whose return value answers two questions at once: whether the toppling happened, and how many grains left the board.

```
ToppleOutcome topple_once(Site x) {
  const Threshold k = mat_.threshold();            // k = 4 = -Delta(x,x)   (2.5b)
  if (h_[idx(x)] < k) return {false, 0};           // stable: T_x is the identity
  h_[idx(x)] -= k;                                 // the diagonal entry
  LossCount loss = 0;
  for (const auto& d : kFourDirections) {
    const auto ni = (long long)x.i + d[0], nj = (long long)x.j + d[1];
    if (ni < 0 || nj < 0 || ni >= n_ || nj >= m_) ++loss;   // row sum < 0 here
    else h_[idx({(std::size_t)ni, (std::size_t)nj})] += 1;  // the +1 entries
  }
  return {true, loss};
}
```

**Listing 2.1:** One toppling. Every line is one entry of row $\boldsymbol{x}$ of (**??**).

The correspondence is item by item: `mat_.threshold()` is $-\Delta(\boldsymbol{x}, \boldsymbol{x})$, the early return is the branch in which $T_{\boldsymbol{x}}$ is the identity, the subtraction is the diagonal entry, and the loop visits exactly the four non-zero off-diagonal entries, incrementing `loss` whenever a neighbour falls outside the board. So `loss` is exactly $-\sum_{\boldsymbol{y}} \Delta(\boldsymbol{x}, \boldsymbol{y})$: the same integer as (**??**), computed the other way round. The two are computed independently and compared, so that a mistake in the boundary test — an unsigned index that wraps around instead of reporting "outside" — cannot corrupt both silently.

## 2.2 Order of toppling, and the stable state

An avalanche is the chain of topplings a drop sets off, and it has an order, because the dynamics says which sites are unstable but never which one to topple next. The rest of this chapter shows that the choice does not matter and isolates which hypothesis does the work.

### 2.2.1 The toppling operator, and commutativity

Write "topple once at $\boldsymbol{x}$" as an operator:

$$(T_{\boldsymbol{x}} p)(\boldsymbol{y}) = \begin{cases} p(\boldsymbol{y}) + \Delta(\boldsymbol{x}, \boldsymbol{y}), & p(\boldsymbol{x}) \geq k, \\ p(\boldsymbol{y}), & p(\boldsymbol{x}) < k. \end{cases} \tag{2.8}$$

It touches $\boldsymbol{x}$ and its four neighbours and acts only when the site should topple; otherwise it is the identity. Toppling $\boldsymbol{x}$ $n$ times adds row $\boldsymbol{x}$ $n$ times, $T_{\boldsymbol{x}}^{n} p = p + n\,\Delta(\boldsymbol{x},\ \cdot\ )$, and a whole avalanche is a product $\mathbb{T} = \prod_i T_{\boldsymbol{x}_i}$ in the order the topplings happened.

**Lemma 2.1.** *Let $\boldsymbol{x}_1 \neq \boldsymbol{x}_2$ satisfy $p(\boldsymbol{x}_1) \geq k$ and $p(\boldsymbol{x}_2) \geq k$. Then both $T_{\boldsymbol{x}_2} T_{\boldsymbol{x}_1} p$ and $T_{\boldsymbol{x}_1} T_{\boldsymbol{x}_2} p$ are defined, and they are equal.*

*Proof.* Put $q = T_{\boldsymbol{x}_1} p$. By (**??**), $\Delta(\boldsymbol{x}_1, \boldsymbol{x}_2) \geq 0$, so $q(\boldsymbol{x}_2) = p(\boldsymbol{x}_2) + \Delta(\boldsymbol{x}_1, \boldsymbol{x}_2) \geq p(\boldsymbol{x}_2) \geq k$: toppling $\boldsymbol{x}_1$ cannot calm $\boldsymbol{x}_2$ down, because a toppling only adds sand to other sites. Hence the branch $q(\boldsymbol{x}_2) \geq k$ applies and $\big(T_{\boldsymbol{x}_2} T_{\boldsymbol{x}_1} p\big)(\boldsymbol{y}) = p(\boldsymbol{y}) + \Delta(\boldsymbol{x}_1, \boldsymbol{y}) + \Delta(\boldsymbol{x}_2, \boldsymbol{y})$. The same computation with the roles exchanged has the same right-hand side, because addition of integers in $\mathbb{Z}^{\boldsymbol{L}}$ is commutative. □

The proof uses $\Delta(\boldsymbol{x}_1, \boldsymbol{x}_2) \geq 0$ and nothing else: not the symmetry half of (**??**), not the row sums, not the number of neighbours, not the shape of the site set. That is why the result survives a change of lattice. It is not the only hypothesis at work, however.

**Remark 2.2.** Equation (**??**) asks two things of a rule: that the instability test at $\boldsymbol{x}$ involves the height of $\boldsymbol{x}$ alone, and that a toppling only adds sand to other sites. The handbook's gradient rule fails the first, since $\boldsymbol{x}$ is unstable when it exceeds a neighbour by $k$ and the trigger therefore involves $p(\boldsymbol{y})$; a rule that may *take* a grain from a neighbour fails the second. Neither property is equivalent to determinism, and random rules are not excluded: the Manna model [**?**], which hands grains to randomly chosen neighbours, is still Abelian [**?**], because the randomness is attached to each site in advance. Randomness drawn *at the moment of toppling* is different, and Dhar's non-Abelian variant [**?**] loses the property that way.

### 2.2.2 Termination and uniqueness

**Proposition 2.3.** *Let $\Delta$ satisfy (**??**)–(**??**) on the finite board $\boldsymbol{L}$. Then every legal sequence of topplings from $p$ has at most $\mathbf{1}^{\top}(-\Delta)^{-1} p$ terms.*

*Proof.* Let $v_t \geq 0$ count the topplings in the first $t$ steps; then $p_t = p + \Delta^{\top} v_t$, and since height is non-negative, $(-\Delta) v_t \leq p$ using the symmetry $\Delta^{\top} = \Delta$. Now $-\Delta$ has positive diagonal by (**??**), non-positive off-diagonal by (**??**) and non-negative row sums by (**??**), strictly positive for at least one row by (**??**), with the off-diagonal pattern of the connected grid graph. It is therefore irreducibly diagonally dominant, hence a nonsingular $M$-matrix [**?**], so $(-\Delta)^{-1} \geq 0$ entrywise and $0 \leq v_t \leq (-\Delta)^{-1} p$ for every $t$. Each toppling increases $\sum_{\boldsymbol{x}} v_t(\boldsymbol{x})$ by one, so the number of steps is bounded by the stated constant, which depends only on the starting configuration. □

**Theorem 2.4** (Abelian lemma)**.** *Let $v$ be the count vector of a legal sequence taking $p$ to a stable configuration, and let $w \geq 0$ be any integer vector with $p + \Delta^{\top} w$ stable. Then $v \leq w$ componentwise.*

*Proof.* Suppose not, and let $t$ be the first step at which some site topples for the $(w(\boldsymbol{x}) + 1)$-st time. Before step $t$ every site has toppled at most $w$ times, so $v_{t-1} \leq w$ with equality at $\boldsymbol{x}$. Step $t$ is legal, so $p_{t-1}(\boldsymbol{x}) \geq k$; but

$$p_{t-1}(\boldsymbol{x}) = p(\boldsymbol{x}) + w(\boldsymbol{x})\Delta(\boldsymbol{x}, \boldsymbol{x}) + \sum_{\boldsymbol{y} \neq \boldsymbol{x}} v_{t-1}(\boldsymbol{y})\,\Delta(\boldsymbol{y}, \boldsymbol{x}) \leq p(\boldsymbol{x}) + \big(\Delta^{\top} w\big)(\boldsymbol{x}) \ \leq\ k - 1, \tag{2.9}$$

since $\Delta(\boldsymbol{y}, \boldsymbol{x}) \geq 0$ for $\boldsymbol{y} \neq \boldsymbol{x}$ lets $v_{t-1} \leq w$ be pushed through the sum term by term, and $p + \Delta^{\top} w$ is stable. That contradiction proves $v \leq w$. □

**Theorem 2.5.** *Every complete legal sequence from $p$ produces the same final configuration and the same toppling-count vector.*

*Proof.* Every sequence is finite by Proposition **??**. For two complete sequences with count vectors $v, v'$, Theorem **??** applied to $v$ with $w = v'$ gives $v \leq v'$, and applied to $v'$ with $w = v$ gives $v' \leq v$. Hence $v = v'$ and the endpoints, being $p + \Delta^{\top} v$, coincide. □

So the stabilisation map $p \mapsto p^{\circ}$ is well defined, an avalanche has a definite size, and the simulation may topple in whatever order is convenient. The count vector is determined by the endpoints alone,

$$p^{\circ} = p + \Delta^{\top} \sigma \qquad \Longleftrightarrow \qquad \sigma = \big(\Delta^{-1}\big)^{\top} \big(p^{\circ} - p\big), \tag{2.10}$$

with no sink vertex or row deletion needed, since (**??**) already makes $\Delta$ invertible. This is the *least action principle* [**?**, **?**]: $\sigma$ is the pointwise smallest non-negative integer vector for which $p + \Delta^{\top} \sigma$ is stable.

### 2.2.3 Four observables, and the program

Once $\sigma$ is known the avalanche is completely described, and the four quantities the handbook asks for [**?**] are four summaries of it:

$$\text{topples } s = \sum_{\boldsymbol{x}} \sigma(\boldsymbol{x}), \qquad \text{area } s_d = \bigl|\{\boldsymbol{x} : \sigma(\boldsymbol{x}) > 0\}\bigr|,$$
$$\text{loss } l = \sum_{\boldsymbol{x}} \sigma(\boldsymbol{x}) \Bigl( - \sum_{\boldsymbol{y}} \Delta(\boldsymbol{x}, \boldsymbol{y}) \Bigr), \qquad \text{length } R = \max_{\boldsymbol{y}:\, \sigma(\boldsymbol{y}) > 0} d(\boldsymbol{x}_0, \boldsymbol{y}). \tag{2.11}$$

The first two count topplings with and without repetition, so $s_d \le s$, with equality exactly when no site topples twice. The loss can only be positive if the avalanche reached the boundary, since an interior toppling has row sum zero. The length is the graph distance from the drop site to the farthest site that toppled; because (**??**) only moves grains along the four grid directions, that distance is the Manhattan distance

$$R = \max_{\boldsymbol{y}:\, \sigma(\boldsymbol{y}) > 0} \bigl( |i_{\boldsymbol{y}} - i_0| + |j_{\boldsymbol{y}} - j_0| \bigr). \tag{2.12}$$

The choice of metric belongs to the geometry, not to the reporting: $R$ counts steps, so the metric must be the one the grains actually walk, and on the hexagonal lattice of Chapter **??** it is not the Manhattan distance.

"Topple until stable" is a queue loop: unstable sites enter, are toppled one at a time, and any site made unstable by a toppling enters in turn.

```
void drop_sand_at(Site s, std::uint32_t n = 1) {
  h_[idx(s)] += n;
  if (h_[idx(s)] >= mat_.threshold()) stabilize(s);   // avalanche
  else history_.push_back(AvalancheStats{});          // no avalanche: zeros
}

AvalancheStats stabilize(Site start) {
  std::deque<Site> q{start};
  std::vector<char> in_q(n_ * m_, 0);                 // dedup on entry
  in_q[idx(start)] = 1;
  ToppleCount s = 0; LossCount loss = 0; std::size_t area = 0, R = 0;
  while (!q.empty()) {
    const Site x = q.front(); q.pop_front(); in_q[idx(x)] = 0;
    if (h_[idx(x)] < k) continue;                     // already absorbed
    loss += topple_once(x).loss; ++s;
    for (const auto& d : kFourDirections) { /* push unstable neighbours */ }
  }
  for (const Site y : fired_list_) R = std::max(R, dist_l1(y, start));
  return {s, area, loss, R};
}
```

**Listing 2.2:** One drop and the relaxation. The order inside the queue is arbitrary by Theorem **??**.

Three details of the loop bear on the theorems. The height of a site is tested again when it leaves the queue, because a neighbour's toppling may have added sand to it while it waited; this is simpler than maintaining the invariant that every queued site is unstable, and it is safe because by (**??**) an unstable site stays unstable until it is toppled. Deduplication on entry changes only the queue length. And replacing the queue by a stack or by a random order changes nothing about the result: that is the Abelian property, and the two-order comparison is the cheapest test of it.

# Chapter 3

# The stationary state

A drop adds one grain and the avalanche that follows removes some; over a long run the board settles into a state in which macroscopic quantities stop drifting while the configuration keeps changing. This chapter describes that state: which configurations it consists of, what the mean height is, and what is known and expected about the distributions of the four avalanche observables of (**??**).

## 3.1 Recurrent configurations and the uniform measure

Not every stable configuration is reachable once the board has been driven for a while. The states the dynamics returns to form the set of *recurrent* configurations $\mathcal{R}$; the rest are transient in the sense that after enough drops they are never visited again. On $\mathcal{R}$ the two ways of sampling a system agree. Sampling *over time* means recording the configurations visited by one long run; sampling an *ensemble* means drawing them independently from the stationary distribution. These differ while the initial condition still influences the run, which is what the burn-in of §**??** removes, and they differ if the chain is not ergodic. Drop-and-relax is irreducible and aperiodic on $\mathcal{R}$ — every recurrent configuration is reachable from every other, and the chain does not cycle — so the stationary distribution there is unique, and the time average along a single run converges to the ensemble average.

That measure is the uniform one. Every recurrent configuration carries the same weight $1/|\mathcal{R}|$, and the number of them is

$$|\mathcal{R}| = \det(-\Delta), \tag{3.1}$$

by the matrix–tree theorem, since $\det(-\Delta)$ counts the spanning trees of the wired graph obtained by attaching a sink to every off-board direction [**?**, **?**, **?**]. The uniformity is what makes the mean height an exactly computable number rather than a fitted one, and it is the reason a measurement of $\langle h \rangle$ is a sharp check of the whole implementation.

The set $\mathcal{R}$ also carries an algebraic structure: the addition operators generate a finite abelian group acting on it, the sand pile group, whose order is $|\mathcal{R}|$ [**?**]. Beyond the count (**??**) nothing in this report depends on that structure, but the group order is computed separately in Chapter **??**.

## 3.2 The mean height

The mean height is the average of the height function over the stationary measure,

$$\langle h \rangle = \frac{1}{L^2} \sum_{\boldsymbol{x}} p(\boldsymbol{x}), \tag{3.2}$$

where the average is taken both over sites and over the ensemble. A first guess is that the four stable heights $0, 1, 2, 3$ are equally likely, giving 1.5. That guess ignores a constraint that the recurrent configurations carry: two neighbouring sites cannot both be empty. The constraint comes from the burning algorithm, in which a site of height zero can be burnt only after all of its neighbours, so that neither of two adjacent empty sites could ever be burnt and the configuration would not be recurrent. Counting the configurations subject to that restriction gives the exact value

$$\langle h \rangle = \frac{17}{8} = 2.125, \tag{3.3}$$

which is already well above the uniform guess.

The individual height probabilities are known exactly as well. Priezzhev computed them for the square lattice [**?**], and the closed forms were later reduced by Jeng, Piroux and Ruelle [**?**] to

$$\begin{aligned} p(0) &= \tfrac{2}{\pi^2} - \tfrac{4}{\pi^3} = 0.0736, & p(1) &= \tfrac{1}{4} - \tfrac{1}{2\pi} - \tfrac{3}{\pi^2} + \tfrac{12}{\pi^3} = 0.1739, \\ p(2) &= \tfrac{3}{8} + \tfrac{1}{\pi} - \tfrac{12}{\pi^3} = 0.3063, & p(3) &= \tfrac{3}{8} - \tfrac{1}{2\pi} + \tfrac{1}{\pi^2} + \tfrac{4}{\pi^3} = 0.4462, \end{aligned} \tag{3.4}$$

and $\sum_i i\, p(i)$ reproduces 2.125; the single integral those formulas contain was later evaluated and proved to take the value that makes this true [**?**, **?**]. The four probabilities and the mean are the sharpest available test of the simulator: they involve no fitting, no cut-off and no assumption about a tail, so a disagreement is a defect rather than a shortage of samples. They also give the exact relation between the threshold and the mean height that Chapter **??** needs, since they show that $\langle h \rangle$ is a property of the lattice and not a constant.

## 3.3 The avalanche distributions

For each avalanche the record holds the four integers of (**??**), and the question is what distribution each of them follows. In the power-law regime the complementary cumulative distribution of a size $X$ behaves as

$$P(X \geq x) \sim x^{-(\tau - 1)}, \tag{3.5}$$

so the slope of the straight part on log–log axes gives $\tau - 1$. The exponent is better estimated by maximum likelihood than by fitting a histogram, since a histogram depends on the binning and the likelihood does not:

$$\hat{\tau} = 1 + \frac{n}{\sum_i \ln(x_i / x_{\min})}, \tag{3.6}$$

where the sum runs over the $n$ events with $x_i \geq x_{\min}$. The two conventions — probability density $x^{-\tau}$ and complementary cumulative $x^{-(\tau-1)}$ — differ by one, and a table of exponents is meaningless unless the convention is stated.

For the toppling number the model has a conjectured exact exponent. The two-dimensional undirected sand pile is equivalent to the $q \to 0$ limit of the Potts model [**?**], and the exponents read off from that equivalence are $\tau_s = 6/5$, $\tau_d = 5/4$ and $\tau_t = 7/5$ for topplings, distinct sites and duration, in the probability-density convention [**?**, **?**]. In the cumulative convention of (**??**) these correspond to slopes 0.2 and 0.25, which is why the distributions are so flat that they can be mistaken for a cut-off over the range a small board can reach.

Two properties of the measurement follow from the finiteness of the board and belong in the analysis from the start. An avalanche cannot exceed the board, so the distribution is cut off at a scale set by $L$, and an estimate of the exponent taken with a lower cut-off near that scale describes the steep edge of the cut-off rather than the power law, returning a value that is too high. The remedy is not a better estimator but a range of sizes: the cumulative distributions of several boards are plotted together, and the exponent is read from a straight section that is common to all of them and lies before the cut. Second, the length $R$ is a geometric quantity bounded by the board and is not expected to be a pure power law at all; it belongs in a finite-size collapse, $s/N^D$ for a cut-off exponent $D$, rather than in a fit of its own. If the collapsed curves fail to overlap, $D$ is the first thing to adjust.

**[TODO]** The exponents $\tau_s$ and $\tau_d$ have not been measured. Doing so requires a run whose usable tail holds on the order of $10^3$ events or more at each of at least three board sizes, so that the collapse described above can be checked before any exponent is quoted.

### 3.4 Correlations between the four observables

The four observables are not independent, and the expected relations follow from their definitions in (**??**).

*Topples against area.* The area counts sites toppled at least once and the toppling number counts all topplings, so $s \geq s_d$, with equality exactly when no site topples twice. In a small avalanche that is the typical case and the two are proportional; in a large avalanche sites near the drop point topple repeatedly, so $s$ grows faster than $s_d$ and one expects $s \sim s_d^{\gamma}$ with $\gamma > 1$. The relation is visible directly as a rising mean of the ratio $s/s_d$ with $s_d$.

*Area against length.* The area of an avalanche is related to its radius by a fractal dimension, $s_d \sim R^{D_f}$. If avalanches filled their bounding region then $D_f$ would be 2; a value below 2 means the cluster is ramified, with a perimeter comparable to the area.

*Loss against length.* This relation is a threshold, not a power law, and it is the sharpest of the four. Loss requires a boundary site to topple, so $l = 0$ for every avalanche that does not reach the boundary; once $R$ exceeds the distance from the drop site to the nearest boundary, loss begins and grows quickly with $R$. A linear correlation coefficient is the wrong summary here, because it averages the exact zeros below the threshold with the growing values above it and reports a weak relation where the relation is in fact deterministic in shape. The appropriate summary is the conditional mean $E[l \mid R]$ after binning by $R$, which reproduces the threshold as a corner at the boundary distance.

*Loss against topples.* Weak, for the same reason: most topplings are interior and conserve sand, so a large avalanche that stays away from the boundary loses nothing.

**[TODO]** The correlation coefficients and the binned conditional mean $E[l \mid R]$ have not been computed. The recorded columns are sufficient for both; the work is post-processing of the avalanche record rather than further simulation.

# Chapter 4

# Data and algorithm

## 4.1 What has to be kept, and how much that is

Two things have to be kept: the current board, $L^2$ heights, and the record of every avalanche, four integers each. Everything else — the toppling matrix, the recurrent-configuration count, the fitted exponents — is either derived on the fly or produced afterwards from the record.

The toppling matrix is the interesting case: it is $N \times N$ with $N = L^2$, but (**??**) gives each row only five non-zero entries, so of the $N^2$ places only about $5N$ carry information.

**Table 4.1:** Size of $\Delta$ against the number of entries it actually uses.

| Board | Sites $N = L^2$ | Places in $\Delta$: $N^2$ | Non-zero |
|---|---|---|---|
| $L = 3$ | 9 | 81 | 37 |
| $L = 50$ | 2 500 | $6.25 \times 10^6$ | $1.25 \times 10^4$ |
| $L = 1000$ | $10^6$ | $10^{12}$ | $5 \times 10^6$ |

At $L = 1000$ the table has a trillion places and five million useful ones. Storing it would be absurd, and it is unnecessary, because $\Delta$ is not data: it is a *function* of the pair $(\boldsymbol{x}, \boldsymbol{y})$, given in closed form by (**??**). Looking up one entry means asking whether $\boldsymbol{x} = \boldsymbol{y}$ and, if not, whether the two sites are neighbours — a constant number of comparisons, so no table is stored and the function is evaluated where it is needed. The five entries of row $\boldsymbol{x}$ are then exactly the diagonal subtraction and the four neighbour increments of `topple_once`, and the two representations are checked against each other so that they cannot drift apart. The dense matrix is formed only for small boards and algebraic checks such as $\det(-\Delta)$ (Appendix **??**).

## 4.2 Storage and the main loop

The board is a one-dimensional array of length $L^2$ flattened in row-major order, $\mathrm{idx}(\boldsymbol{x}) = iL + j$. One index rather than two is a correctness argument rather than a micro-optimisation: toppling touches the four neighbours constantly, and adding one to a single index is easier to verify than changing a pair of indices, especially in the boundary test where an unsigned underflow would silently index the wrong site. Each avalanche appends four integers in order with no summary column, because the later binning, fitting and correlation analysis all need the raw rows.

```
SandPile<BtwMatrix> pile(L, L, seed);
for (std::uint64_t t = 0; t < 2 * L * L; ++t) pile.drop_sand();   // burn-in
pile.clear_history();
std::ofstream out{"avalanches.csv"};
out << "t,s,area,loss,length\n";
for (std::uint64_t t = 1; t <= drops; ++t) {
  pile.drop_sand();
  const AvalancheStats& st = pile.history().back();
  out << t << ',' << st.topples << ',' << st.area << ','
      << st.loss << ',' << st.length << '\n';
}
```

**Listing 4.1:** Main loop: burn-in, sampling, writing the four columns.

Randomness enters in one place only, the uniformly drawn drop site, so the avalanche is deterministic and a fixed seed reproduces a run exactly. That matters for Chapter **??**, where the difference between two lattices must not be a difference between two random streams. The cost of a drop is $O(1)$ plus $O(s)$ for the avalanche, so runtime is proportional to the number of topplings, which is the quantity being measured.

## 4.3 Burn-in

The early part of a run has to be discarded. The board starts empty, so large avalanches are rare while it fills, and including that stage depresses the tail of the distribution and biases any exponent fitted from it upward. The run is therefore driven for $2L^2$ drops and the history cleared before sampling begins.

Whether the board is in the stationary state is checked by quantities that need no fitting and no choice of cut-off: the mean height against the exact value $17/8 = 2.125$ of (**??**), and the mean number of grains lost per drop against 1, which is the input–output balance of §**??**. Both are sensitive to the two errors that would invalidate everything downstream — a wrong boundary test and a wrong threshold — which is why they are checked before any avalanche statistic is recorded.

**[TODO]** The runs whose statistics are reported in Chapter ?? have not yet been carried out. What is needed is a set of runs at three or more board sizes with a common seed policy and a common burn-in, each long enough that the usable tail of the toppling distribution holds on the order of $10^3$ events; the record is then sufficient for both the exponent fitting of §?? and the conditional means of §??.

# Chapter 5

# The hexagonal extension

The handbook's extension 7 replaces the rectangular lattice by a hexagonal one [**?**]. The model has a few quantities that are supposed to be universal and many that are not, and a change of lattice separates them. This chapter defines the extension, shows that it inherits the Abelian property rather than needing a new proof, and compares it with the square baseline quantity by quantity.

## 5.1 Model: a hexagonal lattice

Sites are indexed by axial coordinates $\boldsymbol{x} = (q, r) \in \mathbb{Z}^2$, with six neighbour directions

$$(1,0),\ (0,1),\ (-1,1) \quad \text{and their opposites} \quad (-1,0),\ (0,-1),\ (1,-1). \tag{5.1}$$

Each site therefore has exactly six neighbours, and the tiling whose cells are centred on the sites is the hexagonal one: the cells are regular hexagons and the lattice of cell centres is triangular. The region is a hexagon of radius $R$,

$$\boldsymbol{L}_{\text{hex}} = \{(q,r) : \max(|q|, |r|, |q+r|) \le R\}, \qquad N_{\text{hex}} = |\boldsymbol{L}_{\text{hex}}| = 3R^2 + 3R + 1, \tag{5.2}$$

chosen because its boundary is a single closed ring at constant graph distance $R$ from the centre. There are no 90° corners: the boundary turns through 120° six times.

Two different objects are called "hexagonal". The honeycomb graph — graphene, the hexagonal *tiling* — has three edges per vertex; the lattice used here is the triangular lattice, whose *cells* are hexagons, with six. The model is the six-neighbour one, as the handbook intends, and the distinction matters below: the threshold is fixed by the number of neighbours, so the honeycomb graph would force $k = 3$.

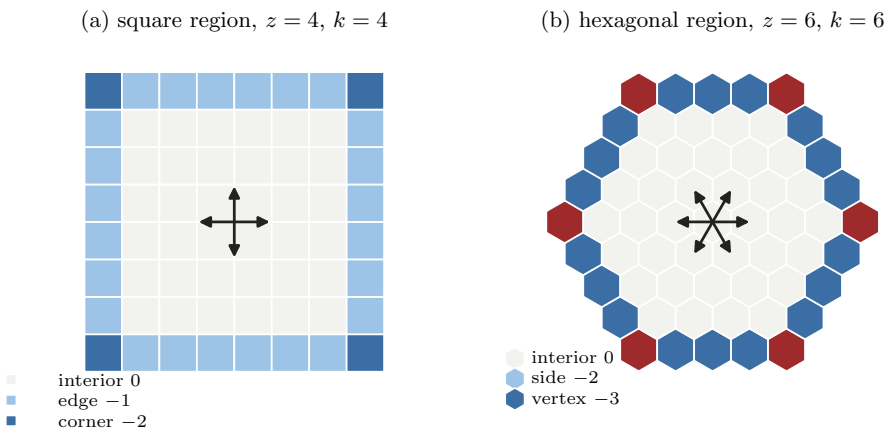



**Figure 5.1:** The two lattices, coloured by the row sum of the toppling matrix, which is the loss incurred by one toppling at that site. Left: the square region, with 0 (interior), −1 (edge) and −2 (corner). Right: the hexagonal region, with 0 (interior), −2 (side) and −3 (vertex). The regions are drawn at matched size — $N = 64$ and $N = 61$ — so that the boundary classes compare directly; the arrows mark the four and six toppling directions.

## 5.2 The threshold is locked by conservation

The toppling matrix is the direct analogue of (**??**):

$$\Delta_{\text{hex}}(\boldsymbol{x}, \boldsymbol{y}) = \begin{cases} -6, & \boldsymbol{x} = \boldsymbol{y}, \\ 1, & \boldsymbol{y} - \boldsymbol{x} \in (??), \\ 0, & \text{otherwise.} \end{cases} \tag{5.3}$$

The four conditions are checked as before, and the check is short enough to write out, because it is where the threshold is decided. (**??**): the direction set (**??**) is closed under negation, so the neighbour relation is symmetric and the off-diagonal entries are all $+1 \ge 0$. (**??**): the diagonal is $-6 < 0$, so

$$k_{\text{hex}} = -\Delta_{\text{hex}}(\boldsymbol{x}, \boldsymbol{x}) = 6. \tag{5.4}$$

(**??**): with $m(\boldsymbol{x})$ the number of the six neighbours lying inside $\boldsymbol{L}_{\text{hex}}$,

$$\sum_{\boldsymbol{y}} \Delta_{\text{hex}}(\boldsymbol{x}, \boldsymbol{y}) = -6 + m(\boldsymbol{x}) \le 0, \tag{5.5}$$

with equality for interior sites and strict inequality at the boundary. (**??**): the region is finite with a boundary, so the total $\sum_{\boldsymbol{x}}(-6+m(\boldsymbol{x})) = -12R-6$ is negative.

The threshold is therefore not a second modelling choice; it is what the matrix says. If the threshold were left at 4 while six grains are sent out, an interior toppling would have row sum +2: each toppling would *create* two grains, condition (**??**) would fail, the total mass would grow without bound, and by Proposition **??** there would be no reason for an avalanche to stop. Conversely, keeping $k=4$ by sending grains to only four of the six neighbours would make the neighbour relation asymmetric, so $\Delta_{\text{hex}}$ would no longer be a graph Laplacian. There is no way to change the degree without changing the threshold.

What that does to the loss is visible in the row sums (Table **??**).

**Table 5.1:** Row-sum classes of $\Delta$ on the two lattices. The hexagonal classes are $\{0,-2,-3\}$: the value $-1$ does not occur, and the deepest class is one grain deeper than on the square lattice.

| Lattice | Class | Row sum | Number of sites |
|---|---|---|---|
| | interior | 0 | $(L-2)^2$ |
| square $L\times L$ | edge | $-1$ | $4(L-2)$ |
| | corner | $-2$ | 4 |
| | interior | 0 | $3R^2-3R+1$ |
| hexagon radius $R$ | side | $-2$ | $6(R-1)$ |
| | vertex | $-3$ | 6 |

Three features of the table matter later. First, the hexagon has three loss classes like the square, but a different three: $\{0,-2,-3\}$ rather than $\{0,-1,-2\}$, since every non-interior site of a hexagonal region has either two or three of its six neighbours outside, so that the transition from interior to boundary is a jump of two grains rather than one. Second, the deepest class is deeper: a hexagon vertex has three directions off the board, against two for a square corner, so there are six points of strongest dissipation rather than four and each of them loses more per toppling. Third, the deepest class carries a larger share of the total loss. The dissipation capacity $\sum_{\boldsymbol{x}}(-\text{row sum})$ is $4L$ on the square and $12R+6$ on the hexagon, of which the deepest class accounts for 8 and 18; at equal $N$ these become $2/\sqrt{N}$ against $3\sqrt{3}/(2\sqrt{N})$, larger by a factor $3\sqrt{3}/4 \approx 1.30$ (Figure **??**c). What is narrower on the hexagon is the relative spread within the boundary: per toppling a square boundary site loses 1 or 2 grains, a factor 2, against 2 or 3, a factor 1.5.

## 5.3 The metric has to change with the geometry

On the square lattice the graph distance is the Manhattan distance, which is why (**??**) is the right definition of $R$: $R$ counts steps. On the hexagonal lattice the shortest path along the six directions is

$$d_{\text{hex}}(\boldsymbol{x},\boldsymbol{y}) = \max\big(|\Delta q|\,,\ |\Delta r|\,,\ |\Delta q+\Delta r|\big), \qquad \Delta q = q_{\boldsymbol{y}}-q_{\boldsymbol{x}},\quad \Delta r = r_{\boldsymbol{y}}-r_{\boldsymbol{x}}, \tag{5.6}$$

and the length observable must be redefined as $R_{\text{hex}} = \max_{\boldsymbol{y}:\,\sigma(\boldsymbol{y})>0} d_{\text{hex}}(\boldsymbol{x}_0,\boldsymbol{y})$.

Keeping the Manhattan distance would be a silent error, and a large one: two sites at hexagonal distance $d$ have axial $L^1$ distance between $d$ and $2d$, since a straight run in one direction and a diagonal run of the same hexagonal length are equally far in the graph metric but differ by a factor of two in $L^1$. Using $L^1$ would inflate measured lengths by up to a factor of two depending on the avalanche's direction, and the threshold relation between $l$ and $R$ of §**??** would be lost while the rest of the code continued to run.

The metric also changes the counting of sites within a radius, which affects finite-size scaling even in a correct implementation. A ball of graph radius $R$ holds $2R^2+2R+1$ sites on the square lattice and $3R^2+3R+1$ on the hexagonal one, a ratio approaching 3/2 (Figure **??**a), so an avalanche of a given radius can involve half again as many sites; and at equal $N$ the hexagonal region has a smaller graph radius, $R \approx \sqrt{N/3}$ against $L=\sqrt{N}$, so the cut-off $s_\xi \sim L^D$ is reached at a different board size. The collapse must therefore be done in the right variable, or the 3/2 will be read as a change of exponent.

## 5.4 Why the hexagonal grid is still Abelian

Three reasons, in increasing order of generality.

*The commutativity proof makes no use of the lattice.* Lemma **??** uses exactly one fact about $\Delta$, namely $\Delta(\boldsymbol{x}_1,\boldsymbol{x}_2) \geq 0$ for $\boldsymbol{x}_1 \neq \boldsymbol{x}_2$, which is (**??**). It does not use the symmetry half of (**??**), the number of neighbours, the shape of the region, or planarity. The same holds for Theorem **??**, where the only place the sign of an off-diagonal entry enters is (**??**), allowing $v_{t-1} \leq w$ to be pushed through the sum. Writing the critical step out for the hexagonal case, with $\boldsymbol{x}_1,\boldsymbol{x}_2$ both unstable and distinct,

$$\big(T_{\boldsymbol{x}_1}p\big)(\boldsymbol{x}_2) = p(\boldsymbol{x}_2) + \Delta_{\text{hex}}(\boldsymbol{x}_1,\boldsymbol{x}_2) \ \geq\ p(\boldsymbol{x}_2) \ \geq\ k_{\text{hex}}, \tag{5.7}$$

so $\boldsymbol{x}_2$ is still unstable and the second toppling happens in either order. The number 6 appears nowhere in this line: it appears only in the diagonal entry, and the diagonal entry does not enter the argument.

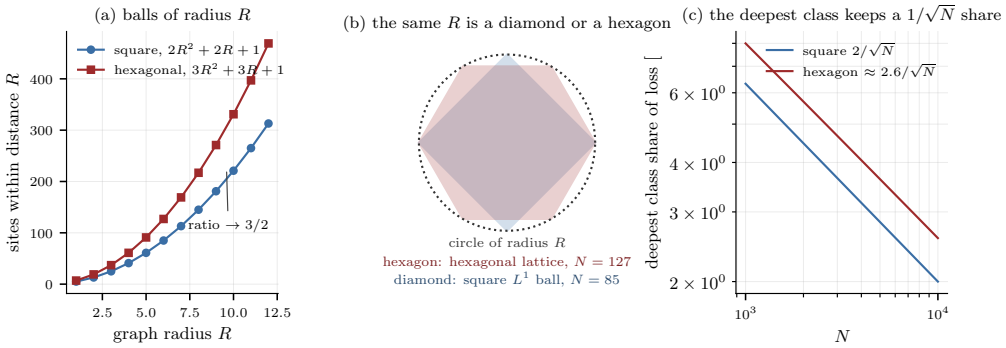



**Figure 5.2:** Comparisons that follow from the two toppling matrices alone and involve no avalanche measurement. (a) Sites within graph distance $R$: $2R^2 + 2R + 1$ on the square lattice against $3R^2 + 3R + 1$ on the hexagonal one, a ratio approaching $3/2$. (b) The two balls at $R = 6$: the square lattice's $L^1$ ball is a diamond and the hexagonal lattice's is a hexagon, neither a circle. (c) The share of the total loss carried by the deepest boundary class as $N \to \infty$: both fall off as $1/\sqrt{N}$, and the hexagon's coefficient is 30% larger.

*Termination and uniqueness transfer for the same reason.* Proposition **??** and Theorem **??** rest on (**??**)–(**??**) through the $M$-matrix property of $-\Delta$, which needs four things of $-\Delta$: non-positive off-diagonal entries, strictly positive diagonal, non-negative row sums with at least one strictly positive, and a connected off-diagonal graph. On the hexagonal region these hold with the substitutions $4 \to 6$ and "square" $\to$ "hexagon": $-\Delta_{\text{hex}}(\boldsymbol{x}, \boldsymbol{y}) = -1 \le 0$ for $\boldsymbol{x} \neq \boldsymbol{y}$; $-\Delta_{\text{hex}}(\boldsymbol{x}, \boldsymbol{x}) = 6 > 0$; row sums $6 - m(\boldsymbol{x}) \ge 0$ by (**??**), strictly positive for each of the $6R$ boundary sites; and the off-diagonal pattern is the adjacency of a connected graph. So $-\Delta_{\text{hex}}$ is again irreducibly diagonally dominant and hence a nonsingular $M$-matrix, $\Delta_{\text{hex}}$ is invertible, and the firing vector

$$\sigma = \left(\Delta_{\text{hex}}^{-1}\right)^{\top}(p^{\circ} - p) \tag{5.8}$$

is unique. Nothing in the argument refers to the lattice beyond those four properties, each of which was checked in §**??** directly from (**??**).

*The general statement, which is the real reason.* The two arguments above are instances of a fact stated at the right level of generality in the literature. Dhar defines the Abelian sand pile on an arbitrary directed graph, with a toppling matrix satisfying only the three sign conditions, and proves the Abelian property at that level [**?**]. Járai defines the same model on an arbitrary finite connected multigraph $G = (V \cup \{s\}, E)$ with a sink, the stability threshold equal to the degree, and $\Delta$ the graph Laplacian restricted to $V$, and proves that the stabilisation map is well defined [**?**]. In both formulations the lattice has already disappeared: it enters only through the degrees and the adjacency, that is, through which entries of $\Delta$ are non-zero. The hexagonal lattice is therefore not a case needing a new proof but an instance of the definition, and the correct statement is conditional: the stabilised configuration is unique *because* $\Delta_{\text{hex}}$ *is a graph Laplacian with non-positive row sums, not because the neighbours happen to be six.*

This identifies what would break the property, and it is not the geometry. Two hypotheses do the work: the instability test must involve the height of the site alone, and a toppling must only add sand to other sites. The hexagonal lattice changes the value of the diagonal entry and which pairs of sites are neighbours, but changes neither hypothesis. Randomness is not by itself a failure either: the Manna model hands grains to randomly chosen neighbours and is still Abelian, because the random rule is fixed per site in advance [**?**, **?**]. Manna's model is Abelian and yet sits in a different universality class, so Abelianity and shared exponents are independent properties — which is exactly the distinction §**??** is built on.

## 5.5 Comparison with the square baseline

The comparison is clearest once the quantities are sorted into those supposed to be universal and those not, because the prediction differs.

### 5.5.1 The exponents are predicted to stay the same

The avalanche exponents are not obtained by fitting them. They follow from an equivalence: the recurrent configurations of the undirected sand pile are in bijection with the spanning trees of the associated graph, and the spanning-tree model is the $q \to 0$ limit of the $q$-state Potts model [**?**, **?**]. In two dimensions that limit is a conformal field theory with central charge $c = -2$, and exponents read off from it depend on $c$, hence on the dimension, and not on the lattice. The chemical-path dimension of spanning trees is $5/4$, giving $(\tau_t - 1) = \frac{4}{5}(\tau_r - 1)$, and the wave decomposition of an avalanche leads to the conjectured exact values

$$\tau_d = \frac{5}{4}, \qquad \tau_s = \frac{6}{5}, \qquad \tau_t = \frac{7}{5} \tag{5.9}$$

**Table 5.2:** The square baseline against the hexagonal extension. Quantities marked *universal* are predicted to agree; those marked *lattice-specific* are predicted to differ.

| Quantity | Square | Hexagonal | Status and reason |
|---|---|---|---|
| degree $z$ / threshold $k$ | 4 | 6 | lattice-specific, forced by (**??**) |
| row-sum classes | $\{0,-1,-2\}$ | $\{0,-2,-3\}$ | lattice-specific, three classes each |
| boundary sites | $4L-4$ | $6R$ | lattice-specific; fraction $\approx 4/\sqrt{N}$ against $\approx 2\sqrt{3}/\sqrt{N}$ |
| total dissipation | $4L$ | $12R+6$ | lattice-specific; per boundary site $\to 1$ against $\to 2$ |
| ball of radius $R$ | $2R^2+2R+1$ | $3R^2+3R+1$ | ratio $\to 3/2$ |
| length metric | $L^1$ | (**??**) | must change with the geometry |
| mean height $\langle h\rangle$ | $17/8 = 2.125$ | $(5+\xi_{\text{hex}})/2$ | lattice-specific; see below |
| toppling exponent $\tau_s$ | 6/5 | 6/5 | universal; see below |
| area exponent $\tau_d$ | 5/4 | 5/4 | universal; same |
| group order $\lvert\mathcal{R}\rvert$ | $\mathrm{e}^{4GN/\pi+o(N)}$ | $\mathrm{e}^{1.62N+o(N)}$ | lattice-specific; see below |

for distinct sites, topplings and duration, in the probability-density convention [**?**]. Because the derivation routes through a two-dimensional conformal field theory and never through the square lattice, the exponents should be the same on the hexagonal lattice: changing the degree from 4 to 6 changes the microscopic coordination, not the dimension.

This is the main falsifiable prediction of the extension. The criterion is fixed in advance: fit $\tau_s$ and $\tau_d$ on the hexagonal lattice at several sizes and require the value to be stable in $N$ and consistent with 6/5 and 5/4 within the fitting error. A deviation that does not shrink as $N$ grows would indicate that the degree affects the effective conservation structure. One caution applies to the comparison. The conjectured values are asymptotic, and a finite board returns a number above them because of the cut-off of §**??**, so the comparison must be between two lattices at the *same* sizes with the *same* procedure, never between a fitted and a theoretical number; and since the cut-off sits at a different place on the two lattices, the size dependence is checked before the exponents are compared.

### 5.5.2 The mean height is predicted to change

The mean height is not universal, and the square baseline's exact value says how and why. On a $z$-regular lattice the mean height and the looping constant are related by

$$\langle h\rangle = \frac{z-1+\xi}{2}, \qquad \xi = \mathbb{E}\big[\#\{\text{neighbours of } o \text{ visited by the infinite LERW from } o\}\big], \tag{5.10}$$

where the loop-erased random walk is obtained by erasing loops from a simple random walk as they are created, and the relation is derived from the burning bijection in [**?**], §5.5.2; see also [**?**] for the looping rate of planar graphs.

The relation explains the square lattice's exact value as a special case. On $\mathbb{Z}^2$ the looping constant is $\xi = 5/4$, proved in [**?**, **?**], so

$$\langle h\rangle_{\text{sq}} = \frac{4-1+5/4}{2} = \frac{17}{8} = 2.125, \tag{5.11}$$

reproducing (**??**) by a different route. It gives the hexagonal prediction in closed form,

$$\langle h\rangle_{\text{hex}} = \frac{6-1+\xi_{\text{hex}}}{2} = \frac{5+\xi_{\text{hex}}}{2}, \tag{5.12}$$

where $\xi_{\text{hex}}$ is the looping constant of the triangular lattice, and it shows that $\langle h\rangle_{\text{hex}}$ is *not* obtained by rescaling 17/8 by the threshold ratio. That argument gives $2.125 \times 6/4 = 3.1875$, which by (**??**) would require $\xi_{\text{hex}} = 1.375$, while keeping $\xi = 1.25$ gives 3.125; the two differ because (**??**) does not fix $\xi_{\text{hex}}$, and no derivation selects one. The relation does give a bound and a target: since $1 \le \xi_{\text{hex}} \le 6$ trivially, $\langle h\rangle_{\text{hex}} \in [3, 5.5]$, and a range comparable to $\xi = 1.25$ narrows this to roughly $[3.1, 3.6]$. It also makes the measurement a sharp one, since measuring $\langle h\rangle_{\text{hex}}$ to four decimals determines $\xi_{\text{hex}}$ to four decimals. On the square lattice the same measurement can be checked against a theorem, because 5/4 is proved; for the triangular lattice no closed form is used here, and the measured value stands on its own.

### 5.5.3 The algebraic complexity changes, without affecting universality

One further quantity is neither a statistic nor an exponent. The number of recurrent configurations is the sand pile group order, $\lvert\mathcal{R}\rvert = \det(-\Delta)$ of (**??**), also the number of spanning trees of the wired graph obtained by attaching a sink to every off-board direction [**?**, **?**]. It grows exponentially in $N$, so the interesting number is the spanning-tree entropy per site, $h = \lim_{N\to\infty} \frac{1}{N} \log \lvert\mathcal{R}\rvert$. On the square lattice this limit is the classical $4G/\pi = 1.16624$, with $G$ Catalan's constant [**?**]; on the hexagonal lattice it must be larger, since each site has more neighbours. Extrapolating $\frac{1}{N}\log\det(-\Delta)$ in $1/R$ gives $h_{\text{hex}} \approx 1.62$, about 39% above the square value, and the same extrapolation on the square lattice returns 1.1672 against the exact 1.16624, an error of $9 \times 10^{-4}$ — which is what makes the hexagonal number

worth quoting to two decimals and no more. Two lattices can therefore share every avalanche exponent and still have sand pile groups of different exponential size: universality is a statement about the statistics, not the algebra, and if $\tau_s$ agrees with the baseline as predicted the group order will still differ by a factor $e^{(1.62-1.166)N}$.

## 5.6 What is to be measured

The program changes in two places only: the toppling rule uses the six directions (**??**), and the length observable uses (**??**). The board is still a one-dimensional array of length $N_{\mathrm{hex}}$, a record is still four integers, and the main loop of §**??** is untouched. That the main loop does not move is itself the check that the geometry was separated properly: the lattice is a parameter object satisfying the `ToppplingMatrix` concept of Appendix **??**, so a new lattice is a new class and not a new simulation. Every quantity below is to be measured with the same seeds, burn-in and fitting code as the baseline, and the board size reported alongside each number, since both the exponent and the cut-off $s_\xi \sim L^D$ depend on it.

**Table 5.3:** Quantities to be measured on the hexagonal lattice, with the expected value and the criterion that would count as a disagreement.

| **Quantity** | **Expected** | **Test and criterion** |
|---|---|---|
| threshold $k_{\mathrm{hex}}$ | 6 | read from $-\Delta_{\mathrm{hex}}(\boldsymbol{x},\boldsymbol{x})$; a different value would mean (**??**) fails |
| mean height $\langle h\rangle_{\mathrm{hex}}$ | $(5+\xi_{\mathrm{hex}})/2$ | average after $2N$ burn-in; the measured value determines $\xi_{\mathrm{hex}}$ |
| loss rate $\bar{l}$ | 1 | grains lost per drop; input–output balance, must hold on both lattices |
| topples $\tau_s$, area $\tau_d$ | 6/5 and 5/4 | MLE at a common $x_{\min}$ and common $N$; agreement within fitting error accepts one universality class, a size-stable deviation reports new exponents |
| length $\tau_R$, $E[l \mid R]$, $D$ | metric-dependent | measure $\tau_R$ with (**??**); bin $E[l \mid R]$ to expose the corner; at least three sizes, and if the $s/N^D$ curves do not overlap adjust $D$ first |
| boundary row sums | $\{0,-2,-3\}$ | recomputed from $\Delta_{\mathrm{hex}}$; a value of $-1$ would mean the region is not a hexagon |
| Abelian property | holds | stabilise one configuration by queue, stack and random order; the firing vectors must agree exactly |

**[TODO]** The measurements of Table **??** have not been carried out. The implementation still to be added is a second class satisfying the `ToppplingMatrix` concept, with the directions (**??**) and the metric (**??**); after that the simulation is a recompilation of the existing code. The order of work is: first the square-lattice runs of §**??**, long enough to fix $\tau_s$ and $\tau_d$ on the baseline; then the same runs on the hexagonal lattice at matched sizes; and finally the collapse across sizes, with $D$ adjusted before any exponent is reinterpreted. The criterion for accepting a single universality class is stated with the exponents above.

# Chapter 6

# Conclusions

## 6.1 What the model turned out to be

The BTW sand pile is usually introduced as a rule about four grains and four neighbours. The useful object is not the rule but the table it defines. Once the rule is written as the toppling matrix $\Delta$ of (**??**), four separate claims — that topplings commute, that an avalanche ends, that its endpoint is unique, and that the number of recurrent configurations is $\det(-\Delta)$ — all follow from four sign conditions on that table, each constraining one part of the dynamics. The chain runs (**??**) $\Rightarrow$ commutativity, (**??**) and (**??**) $\Rightarrow$ termination and uniqueness, and (**??**) and (**??**) $\Rightarrow$ the graph-Laplacian reading that makes the algebra possible. Nothing in the chain refers to squares.

That reframing is what made the extension cheap. The hexagonal lattice of Chapter **??** needs a new matrix, threshold and distance, but no new proof of the Abelian property, because the existing proof never uses the lattice. The general statements in the literature — the model on an arbitrary directed graph [**?**] and on an arbitrary finite multigraph with the degree as threshold [**?**] — say the same at the level of generality where the question does not arise. The corresponding rule for later work is that two things must be checked, and neither is geometric: the instability test must involve the height of the site alone, and a toppling must never remove sand from another site. Changes of geometry, degree, boundary shape or dimension preserve both; changes of *rule* may not. Randomness alone is not the problem — the Manna model [**?**] is still Abelian [**?**] — which shows that Abelianity and shared exponents are separate properties.

The stationary state is correspondingly well characterised. The recurrent configurations carry a uniform measure with weight $1/|\mathcal{R}|$, where $|\mathcal{R}| = \det(-\Delta)$ counts the spanning trees of the wired graph, and that measure fixes the mean height exactly: $\langle h \rangle = 17/8$ on the square lattice, with the four height probabilities known in closed form. Those are the quantities against which any implementation is checked, because they involve no fitting, no cut-off and no assumption about a tail.

## 6.2 What the hexagonal comparison settled

The comparison was carried out for everything that follows from the two toppling matrices alone, and it splits cleanly. *Universal*: the exponents $\tau_s = 6/5$ and $\tau_d = 5/4$, since their derivation routes through the $q \to 0$ Potts model and the two-dimensional conformal field theory with $c = -2$ [**?**, **?**], which is sensitive to the dimension and not to the coordination number. *Lattice-specific and derivable*: the threshold is forced to 6 by (**??**); the row-sum classes become $\{0, -2, -3\}$; the mean height becomes $(5 + \xi_{\text{hex}})/2$, with $17/8$ the case $\xi = 5/4$; and the sand pile group grows at a different exponential rate, $h_{\text{hex}} \approx 1.62$ against $4G/\pi = 1.16624$. *Lattice-specific and affecting the analysis rather than the physics*: the 3/2 ball-size ratio puts the finite-size cut-off elsewhere, so the collapse variable must be chosen with care and the two lattices must be compared at matched sizes.

Two features of the change of threshold follow directly from the row sums and are easy to get wrong. A hexagonal boundary is deeper, not shallower: its six vertices lose three grains per toppling against two for a square corner, and at equal $N$ the deepest class carries 30% more of the total loss. Only the relative spread of the per-toppling loss is narrower. What the hexagonal boundary does have is isotropy — constant graph distance from the centre in all six directions, where a square boundary is at $L/2$ along an axis and $L/\sqrt{2}$ along a diagonal — and isotropy, not evenness of loss, is the property that bears on the shape of large avalanches and on the cut-off.

The measurements that would test the universal-exponent prediction are listed in Table **??**. The single most informative number is the mean height, since by (**??**) it determines the looping constant of the triangular lattice.

# Appendix A

# Types, signatures and the matrix layer

## A.1 Core types and function signatures

The type layer is organised so that an illegal state cannot be represented. The clearest instance is the split between `Height` and `DeltaVal`: a number of grains cannot be negative, and a matrix entry must be able to be, because (**??**) requires $\Delta(\boldsymbol{x}, \boldsymbol{x}) < 0$. Making that difference a property of the type means "$-1$ grain" and "a positive diagonal entry" cannot be written down at all, and the compiler reports the mistake instead of the simulation producing wrong statistics.

```
using Height    = std::uint32_t;   // grains on a site; (2.1); negative counts unrepresentable
using DeltaVal  = std::int32_t;    // Delta(x, y); the only signed quantity, because of (2.5b)
using Threshold = std::uint32_t;   // k = -Delta(x, x) = 4
using ToppleCount = std::uint64_t; // a large avalanche can exceed 32 bits
using LossCount   = std::uint64_t;

struct Site { std::size_t i{}, j{}; constexpr bool operator==(const Site&) const = default; };

struct ToppleOutcome {
  bool      fired = false;   // did T_x topple, or act as the identity?      (2.9)
  LossCount loss  = 0;       // grains leaving the system = -(row sum),      (2.4)
};

struct AvalancheStats {
  ToppleCount topples{};   // s   : total topplings, counting repeats      (2.15)
  std::size_t area{};      // s_d : distinct sites toppled, s_d <= s       (2.15)
  LossCount   loss{};      // l   : grains lost at the boundary            (2.15)
  std::size_t length{};    // R   : graph radius from the drop site        (2.16)
};

template <typename M>
concept ToppingMatrix = requires(const M m, Site x, Site y) {
  { m.threshold() }  -> std::convertible_to<Threshold>;
  { m.delta(x, y) }  -> std::convertible_to<DeltaVal>;   // (2.3)
  { m.row_sum(x) }   -> std::convertible_to<DeltaVal>;   // (2.4)
  { m.verify_all() } -> std::convertible_to<bool>;       // (2.5a)-(2.5d)
};

class BtwMatrix {                       // the square lattice, z = 4, k = 4
 public:
  DeltaVal  delta(Site x, Site y) const noexcept;    // the pure function of (2.3)
  DeltaVal  row_sum(Site x) const;                   // 0 / -1 / -2
  bool      verify_all() const;                      // (2.5a)-(2.5d), small boards
  Eigen::MatrixXi dense() const;                     // small boards only, for det(-Delta)
};

template <ToppingMatrix Matrix = BtwMatrix>
class SandPile {
 public:
  SandPile(std::size_t n, std::size_t m, std::uint64_t seed = 42);
  void      drop_sand(std::uint32_t n = 1);          // uniform random site
  ToppleOutcome topple_once(Site x);                 // one application of T_x, (2.9)
  AvalancheStats stabilize(Site start);              // the queue loop of 2.9
  double    mean_height() const;                     // <h> = mass / (N M)
  void      clear_history();                         // discard the burn-in
  bool operator==(const SandPile& o) const;          // compare two toppling orders
};
```

**Listing A.1:** Basic types, the toppling matrix contract, and the sand pile class. The comments are the mathematical definitions.

The `ToppingMatrix` concept is the contract that makes the lattice a parameter: a class satisfies it if it can be constructed from a size, reports its threshold, answers $\Delta(\boldsymbol{x}, \boldsymbol{y})$ and the row sum, and verifies the four conditions on small boards. A hexagonal lattice is a second class satisfying the same concept, and `SandPile` is instantiated on either, so Chapter **??** is a recompilation rather than a second simulator. Two further details are load-bearing rather than stylistic: `DeltaVal` being signed lets one function return both the $+1$ of a neighbour increment and the $-4$ of the diagonal subtraction; and `operator==` on `SandPile` exists so that the same configuration can be stabilised in two orders and the boards compared site by site, which is the executable form of Theorem **??**.

## A.2 The matrix layer

`BtwMatrix` stores only its two dimensions. It does not store $\Delta$, because $\Delta$ is a closed-form function of the pair $(\boldsymbol{x}, \boldsymbol{y})$ and §**??** is the argument that materialising it is impossible at $L = 1000$. The lookups are therefore the definitions: `delta(x, y)` is (**??**) verbatim, with asserted preconditions that compile away in a release build; `row_sum(x)` is (**??**) and is the only place the board boundary appears; `check_4a()` through `check_4d()` are (**??**)–(**??**) as executable predicates at cost $O(N^2)$, used on small boards; and `dense()` exports $\Delta$ as an `Eigen::MatrixXi` for checks needing a real matrix, such as $\det(-\Delta) = 192$ on a $2 \times 2$ board, which is the one third-party dependency and a compile-time-only one. A `static_assert(ToppingMatrix<BtwMatrix>)` at the end of the header pins the contract down, so that a later change to the signature fails the build rather than letting a simulation use the wrong rule.